

Cypher-Lock Preamble for Asynchronously Sampled Data Channels

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The latest semiconductor technology nodes allow for 4X oversampling rates in HDD data channels. This enables a new asynchronous sampling architecture where the oversampled signal is interpolated to produce synchronous sampling. This contrasts with the dominant production architecture where the sampling clock is interpolated to produce synchronous sampling. With this new asynchronous architecture, a new preamble strategy referred to as ‘Cypher-Lock’ takes advantage of communication techniques to bring advancements to the sector format for HDD.

Index Terms— Magnetic Recording, Synchronous Sampling, Timing & Gain Acquisition, Iterative Decoding, Partial-Response Maximum-Likelihood (PRML), and Low-Density Parity-Check (LDPC).

I. INTRODUCTION

THE history of the HDD sampled signal data channel goes back to 1990 when the first production Partial-Response Maximum-Likelihood data channel was shipped in the IBM 0681 “Redwing” hard disk drive [1]. This feat was enabled by IBM 1.0 μ m mixed signal IC technology with an encoded baud rate of 27 MHz. From this starting point, up to the mid 2010’s, data channels were largely constrained to baud rate sampling because of Integrated Circuit (IC) technology limits.

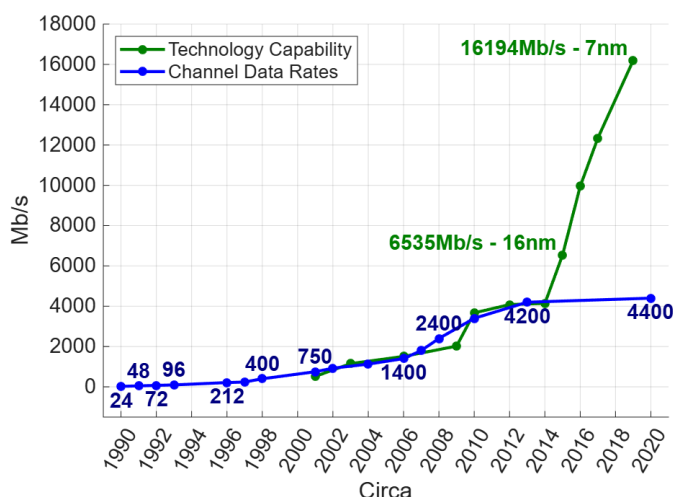


Fig. 1. HDD Channel Data rate vs Technology Node Speed

Figure 1 illustrates some historical information of channel data rates vs the measured ring speed of IC technologies in use at that time. As can be interpreted from the chart, data channel rates have remained quite flat over the last decade, whereas IC technology has steadily progressed in speed. This divergence has brought an opportunity for oversampling of the signal in the data channel. Oversampling of the signal allows for accurate and easy time interpolation of the signal. Time interpolation to 1/64 of a bit resolution is a typical requirement and can be achieved with an oversampled signal. Cypher-Lock, presented here, is a rethinking of the sector acquisition process and reimagines the efficiency by which sampling synchronization and framing can be achieved.

II. CYPHER-LOCK SCHEME

The conventional HDD data channel sector format includes a two-stage preamble. The first stage bit sync field is a constant magnet pattern used for gain and phase adjustment to achieve synchronous sampling. The second stage sync word pattern is used to determine the Start of Data (SOD). The process flow of the sync word detection is usually by equalization of the synchronous samples, Viterbi trellis bit detection, then some type of robust pattern recognition algorithm. The algorithm must be robust to small defects and capable of operating at very poor Signal-to-Noise Ratio (SNR).

The Cypher-Lock scheme, as shown in Figure 2, utilizes a single integrated preamble field. It serves as gain and phase measurement to achieve synchronous sampling. It also provides a robust correlation method used to identify the SOD. It does this using only asynchronous samples and without the need for equalization of the sampled signal and without the need for a Viterbi trellis detector.

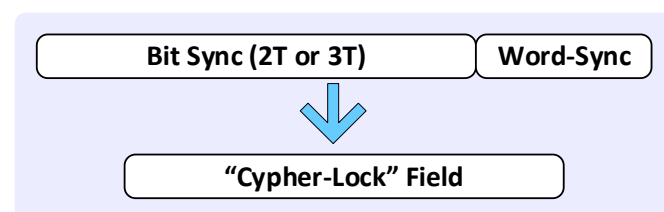


Fig. 2. Single Integrated Cypher-Lock Field

III. PHASE-MODULATED CARRIER

Cypher-Lock introduces a well-known communication technique known as Phase Modulation [2]. A carrier tone is generated that is used to determine gain and phase information and that later allows the signal samples to be interpolated to obtain synchronous intra-bit channel operation. The carrier is also phase modulated with a Pseudo-Random Binary-Sequence (PRBS) that allows the SOD inter-bit location to be uniquely identified. PRBS has a unique property that it correlates with itself only once per cyclic period.

Figure 3 illustrates one full period of an example Cypher-Lock field. In this example, the carrier tone has a frequency of

$1/(6T)$, where T is an encoded baud rate magnet length. A PRBS sequence of length $2^4-1=15$ is used to modulate the carrier. The modulation of the carrier is performed by phase shifting the rising edges of the carrier tone. A logical 1 of the PRBS shifts the rising edge $1T$ to the left. A logical 0 of the PRBS shifts the rising edge $1T$ to the right. One full Cypher-Lock period contains 15 modulated cycles of the carrier for a total length of $6T*15=90T$.

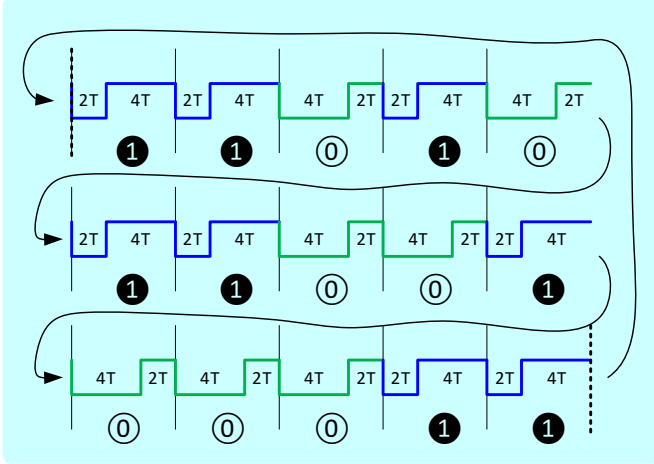


Fig. 3. Example Full Cypher-Lock Period

IV. CYPHER-LOCK CORRELATORS

Three main correlator sequences are used to extract the information in the phase modulated signal. These are illustrated in Figure 4.

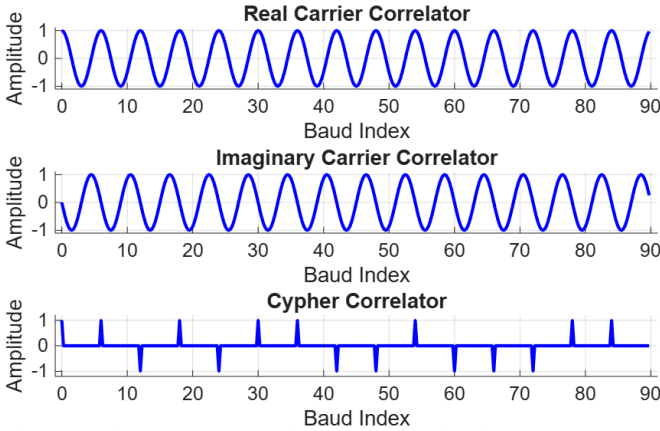


Fig. 4. Correlator Sequences used with Cypher-Lock Scheme

The real and imaginary carrier correlators are used to extract the gain and timing information from the carrier using conversion to magnitude and phase format. The cypher correlator is used in multiple circularly shifted versions to locate the SOD in a region with variation (e.g., ± 20 bits). For instance, since the Cypher-Lock pattern is periodic, let us say that 130 total preamble bits are written to the disk. This would correspond to approximately 1.44 periods of the Cypher-Lock pattern. The evaluation of the Cypher-Lock field is performed on exactly a $90T$ capture length of the samples. The capture window is allowed to be anywhere within the $130T$ written

preamble area. The excess written pattern length is what provides robustness to uncertainty in the preamble location.

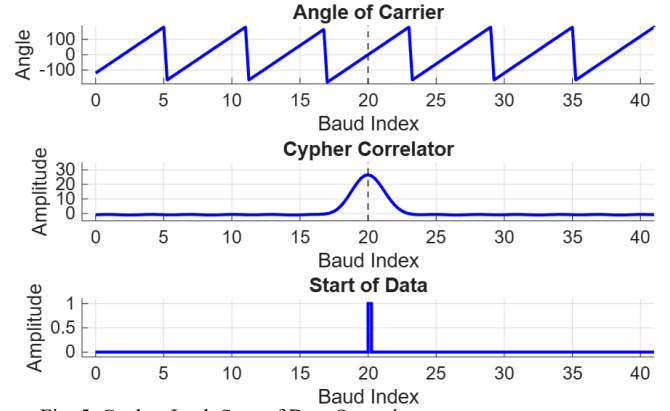


Fig. 5. Cypher-Lock Start of Data Operation

Figure 5 demonstrates how the Cypher-Lock SOD is found. By definition, the inter-bit position of the SOD is where the carrier correlation angle rises through zero. The bank of cypher correlators are used as a qualifier. Together, the carrier angle and maximum cypher correlation uniquely define the time position of the SOD.

V. CYPHER-LOCK PERFORMANCE

The performance of the Cypher-Lock preamble is required to be significantly better than the Sector Failure Rate (SFR) of the data portion of the sector. The ability to return valid data to the customer, which can include many Data Recovery Procedures (DRP), should concentrate on ECC limitations and not on a lack of robustness of the sector acquisition process. As can be seen in Figure 6, The Cypher-Lock scheme can achieve over 22dB better performance compared to the iterative decoding performance of the data portion of the sector. This result helps confirm the capability of the Cypher-Lock approach.

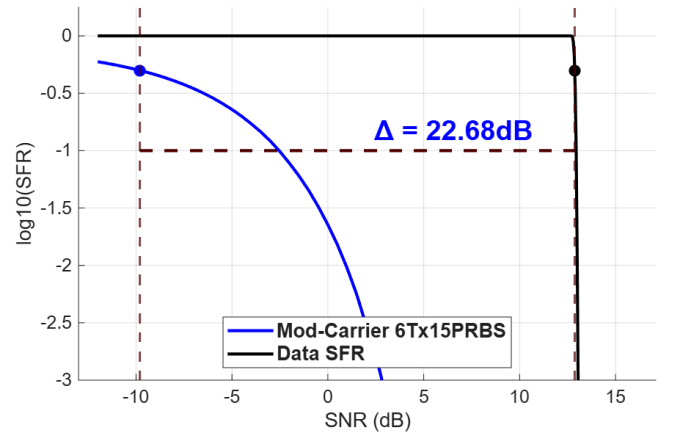


Fig. 6. Cypher-Lock Synthetic Performance

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